

you can think of x_i as fixed characteristics which do not change across different samples of a given population, non-randomness means that we can directly take unconditional expectations/we don't need to condition on x

$u_{i,e} \sim \mathcal{N}(0, \sigma_u^2)$ i.e. $\sum_{i=1}^n m_i$ employees total

* What does it mean to assume $x_{i,e}$ is non-random, and why do we do that here? *

$$\bar{y}_i = \beta_0 + \beta_1 \bar{x}_i + \bar{u}_i \quad \text{with} \quad \bar{u}_i = \frac{1}{m_i} \left(\sum_{e=1}^{m_i} u_{i,e} \right)$$

$$\begin{aligned} \text{a) } E[\bar{u}_i] &= \frac{1}{m_i} \sum_{e=1}^{m_i} E[u_{i,e}] \quad \text{by linearity} \\ &= \frac{1}{m_i} \sum_{e=1}^{m_i} 0 \quad \text{since } u_{i,e} \text{ is mean zero across } i \text{ and } e \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{var}(\bar{u}_i) &= \frac{1}{(m_i)^2} \text{var} \left(\sum_{e=1}^{m_i} u_{i,e} \right) \quad \text{by var rules} \\ &= \frac{1}{m_i} \sum_{e=1}^{m_i} \text{var}(u_{i,e}) \quad \text{as each } u_{i,e} \text{ is independent} \\ &= \frac{1}{m_i} \sigma_u^2 \quad \text{as since each has variance } \sigma_u^2 \end{aligned}$$

$$\begin{aligned} \text{cov}(\bar{u}_i, \bar{u}_j) &= E[\bar{u}_i \bar{u}_j] \quad \text{as both mean zero so } E[\bar{u}_i]E[\bar{u}_j] = 0 \\ &= E \left[\left(\frac{1}{m_i} \sum_{e=1}^{m_i} u_{i,e} \right) \left(\frac{1}{m_j} \sum_{f=1}^{m_j} u_{j,f} \right) \right] \end{aligned}$$

1a. yes, correct! $= \frac{1}{m_i m_j} E \left[\sum_{e=1}^{m_i} u_{i,e} \cdot \sum_{f=1}^{m_j} u_{j,f} \right]$

$$= \frac{1}{m_i m_j} \sum_{e=1}^{m_i} \sum_{f=1}^{m_j} E[u_{i,e} \cdot u_{j,f}] \quad \text{by linearity}$$

but since $i \neq j$ and $\{u_{i,e}\}$ are independent, $u_{i,e}$ and $u_{j,f}$ are uncorrelated $\forall e, f$. So (given they're each mean zero), the expectation of their product is zero (by argument above)

$$\therefore = \frac{1}{m_i m_j} \sum_{e=1}^{m_i} \sum_{f=1}^{m_j} 0 = 0$$

b) $(\hat{\beta}_0, \hat{\beta}_1)$ solves $\min_{(b_0, b_1) \in \mathbb{R}^2} \sum_{i=1}^n (\bar{y}_i - b_0 - b_1 \bar{x}_i)^2$, and we have

that $\hat{\beta}_1 = \frac{n^{-1} \sum (\bar{x}_i - \bar{\bar{x}})(\bar{y}_i - \bar{\bar{y}})}{n^{-1} \sum (\bar{x}_i - \bar{\bar{x}})^2}$ where $\bar{\bar{x}} := \frac{1}{n} \sum_{i=1}^n \bar{x}_i$

Substituting in the model, $\hat{\beta}_1 = \beta_1 + \frac{\sum (\bar{x}_i - \bar{\bar{x}}) \bar{u}_i}{\sum (\bar{x}_i - \bar{\bar{x}})^2}$ * When answering

I would say yes but should check with the tutor! exam questions, is showing the derivation here required? *

So $E[\hat{\beta}_1] = \beta_1 + E\left[\frac{\sum (\bar{x}_i - \bar{\bar{x}}) \bar{u}_i}{\sum (\bar{x}_i - \bar{\bar{x}})^2}\right]$ by linearity

~~We can apply the LIE and condition on the $\bar{x}_i$'s, i.e. $E[\hat{\beta}_1 | \bar{x}_i]$~~ consider

~~By conditioning, we can write the numerator as~~

~~Since the $\bar{x}_i$'s are non-random, we can pull them out of the expectation, and we then have~~

$$= \beta_1 + \frac{\sum (\bar{x}_i - \bar{\bar{x}})}{\sum (\bar{x}_i - \bar{\bar{x}})^2} \underbrace{E[\bar{u}_i]}_{=0 \text{ by assumption}}$$

$= \beta_1$ provided there is some variation in $\bar{x}_i$ so the denominator is nonzero

1b. yes, correct!

$$\begin{aligned} \text{var}(\hat{\beta}_1) &= \text{var}\left(\frac{\sum (\bar{x}_i - \bar{\bar{x}}) \bar{u}_i}{\sum (\bar{x}_i - \bar{\bar{x}})^2}\right) = \underbrace{\frac{(\sum (\bar{x}_i - \bar{\bar{x}})^2)^2}{(\sum (\bar{x}_i - \bar{\bar{x}})^2)^2}}_{=: L} \text{var}(\bar{u}_i) \quad \text{Since each } \bar{u}_i \text{ is zero covariate with } \bar{u}_j, i \neq j \text{ by (a)} \\ &= L \cdot \frac{1}{n_i} \sigma_u^2, \text{ by (a)} \end{aligned}$$

$$\begin{aligned} c) E[\bar{u}_i^*] &= E\left[\left(\frac{1}{m_i} \sum u_{i,e}\right) / \sqrt{m_i}\right] = m_i^{-1} \cdot (m_i^{-1})^{-1/2} \sum E[u_{i,e}] \quad \text{by linearity} \\ &= 0 \text{ as in (a).} \end{aligned}$$

1c. ok, correct!

$$\begin{aligned} \text{var}(\bar{u}^*) &= \text{var}\left(\frac{1}{m_i} \sum u_{i,e}\right) = \text{var}(\bar{u}_i \cdot m_i^{1/2}) = m_i \text{var}(\bar{u}_i) \\ &= \sigma_u^2 \quad \text{by (a)} \end{aligned}$$

$$\text{Cov}(\bar{u}_i^*, \bar{u}_j^*) = \frac{1}{m_i^{1/2} m_j^{1/2}} \text{Cov}(\bar{u}_i, \bar{u}_j) = 0 \text{ as in (a)}$$

d) As usual, $\tilde{\beta}_0 = \frac{1}{n} \sum \bar{y}_i / \sqrt{h_i}$ and $\tilde{\beta}_1 = \frac{1}{n} \sum \bar{x}_i / \sqrt{h_i}$
 and $\tilde{\beta}_1 = \frac{\sum (\bar{x}_i / \sqrt{h_i} - \bar{\bar{x}} / \sqrt{h_i})(\bar{y}_i / \sqrt{h_i} - \bar{\bar{y}} / \sqrt{h_i})}{\sum (\bar{x}_i / \sqrt{h_i} - \bar{\bar{x}} / \sqrt{h_i})^2}$

The estimators $(\tilde{\beta}_0, \tilde{\beta}_1) = \underset{(b_0, b_1) \in \mathbb{R}^2}{\operatorname{argmin}} \sum_{i=1}^n (\bar{y}_i^* - b_0^* - b_1^* \bar{x}_i^*)^2$

where $z_i^* := z_i / \sqrt{h_i} \equiv z_i m_i^{1/2}$

We want the standard FOCs $\frac{\partial L}{\partial b_0} \stackrel{!}{=} 0$; $\frac{\partial L}{\partial b_1} \stackrel{!}{=} 0$.
 (1) (2)

So we have

(1) $-2 \sum (\bar{y}_i^* - b_0^* - b_1^* \bar{x}_i^*) = 0 \Rightarrow \sum (\bar{y}_i - b_0 - b_1 \bar{x}_i)$
 (2) $-2 \sum (\bar{y}_i^* - b_0^* - b_1^* \bar{x}_i^*) \bar{x}_i^* = 0$

d) The estimators $(\tilde{\beta}_0, \tilde{\beta}_1) = \underset{(b_0, b_1) \in \mathbb{R}^2}{\operatorname{argmin}} \sum_{i=1}^n [(\bar{y}_i - b_0 - b_1 \bar{x}_i) \sqrt{m_i}]^2$

Since $h^{-1/2} = (m^{-1})^{-1/2} = m^{1/2}$

We have FOCs (1) $\frac{\partial L}{\partial b_0} = 0$ and (2) $\frac{\partial L}{\partial b_1} = 0$.

(1) $-2 \sum (\bar{y}_i - b_0 - b_1 \bar{x}_i) m_i = 0$
 (2) $-2 \sum (\bar{y}_i - b_0 - b_1 \bar{x}_i) m_i \bar{x}_i = 0$ } Normal eq's solved by $(\tilde{\beta}_0, \tilde{\beta}_1)$

Rearranging (1), we have the familiar expressions:

$\tilde{\beta}_0 = \frac{1}{N} \sum \bar{y}_i m_i - \tilde{\beta}_1 \frac{1}{N} \sum \bar{x}_i m_i$

where $N = \sum_{i=1}^n m_i$
 is total no. of employees.

~~$\tilde{\beta}_1 = \frac{\sum m_i \bar{x}_i \bar{y}_i - \frac{1}{N} \sum m_i \bar{x}_i \sum m_i \bar{y}_i}{\sum m_i \bar{x}_i^2 - \frac{1}{N} (\sum m_i \bar{x}_i)^2}$~~

d) To find $\tilde{\beta}_1$, we can rearrange (2) and substitute in:

$$(\sum m_i \bar{x}_i^2) \cdot \tilde{\beta}_1 = \frac{\sum m_i \bar{y}_i \bar{x}_i - \tilde{\beta}_1 \sum m_i \bar{x}_i}{\sum m_i \bar{x}_i^2}$$

1d. ok, correct!

$$\left(\sum m_i \bar{x}_i^2 - \frac{1}{N} \sum \bar{x}_i m_i \sum \bar{x}_i m_i \right) \tilde{\beta}_1 = \sum m_i \bar{y}_i \bar{x}_i - \left(\frac{1}{N} \sum \bar{y}_i m_i \right) \left(\sum m_i \bar{x}_i \right)$$

$$\therefore \tilde{\beta}_1 = \frac{\sum m_i \bar{y}_i \bar{x}_i - \frac{1}{N} \sum \bar{y}_i m_i \sum m_i \bar{x}_i}{\sum m_i \bar{x}_i^2 - \frac{1}{N} \left(\sum \bar{x}_i m_i \right)^2}$$

which does look similar-ish to the usual OLS formula, except now we weight each firm-level aggregate by its number of employees. **yes, exactly**

~~If we let $\bar{x}_w$ be the weighted average of company-level averages so $\bar{x}_w = \frac{1}{N} \sum \bar{x}_i m_i$, we can write~~

$$\tilde{\beta}_1 = \frac{\sum m_i \bar{y}_i \bar{x}_i - \frac{1}{N} \sum \bar{x}_i m_i \sum \bar{y}_i m_i}{\sum m_i \bar{x}_i^2 - \frac{1}{N} \left(\sum \bar{x}_i m_i \right)^2}$$

e) Note that $\tilde{\beta}_1$ is unbiased, just as $\hat{\beta}_1$ is. Like in b), we can substitute in the model and solve to obtain

1e. yes, correct!

$$\tilde{\beta}_1 = \beta_1 + \frac{\sum (\bar{x}_i^* - \bar{x}^*) \bar{u}_i^*}{\sum (\bar{x}_i^* - \bar{x}^*)^2}, \text{ and similarly}$$

use eq conditioning to show $E[\tilde{\beta}_1] = \beta_1$.

Also, notice that model (2) has homoskedastic errors: from (c), we have $\text{var}(\bar{u}^*) = \sigma_u^2$. Since our errors are (i) conditional mean zero and independent of regressors, (ii) have finite conditional variance σ_u^2 , and (iii) are uncorrelated, we can apply the Gauss-Markov theorem. So, $\tilde{\beta}_1$ is BLUE, while $\hat{\beta}_1$ (with heteroskedastic errors) is not. Thus, we should use $\tilde{\beta}_1$.

2. $y_i = \beta + u_i$, $u_i \sim N(0, (1/2)^i)$

a) $\hat{\beta} = \underset{b \in \mathbb{R}}{\operatorname{argmin}} \sum_{i=1}^n (y_i - b)^2$ which is solved by the sample mean, from our FOC $\frac{\partial L}{\partial b} = -2 \sum (y_i - b) \stackrel{!}{=} 0$

i.e. $\hat{\beta} = \bar{y} := \frac{1}{n} \sum y_i = \frac{1}{n} \sum (\beta + u_i)$
 $= \beta + \frac{1}{n} \sum u_i$

So $E[\hat{\beta}] = \beta + E[\frac{1}{n} \sum u_i] = \beta + \frac{1}{n} \sum E[u_i]$ by linearity
 $= \beta + \frac{1}{n} \sum 0 = \beta$ since each u_i is mean zero.
 $\therefore \hat{\beta}$ is unbiased.

$\operatorname{var}(\hat{\beta}) = \operatorname{var}\left(\frac{1}{n} \sum u_i\right) = \frac{1}{n^2} \sum \operatorname{var}(u_i)$ since each $u_i \perp u_j$, $i \neq j$
 $= \frac{1}{n^2} (0.5 + 0.5^2 + \dots + 0.5^n)$

$= \frac{1}{n^2} \cdot \frac{0.5(0.5^n - 1)}{0.5 - 1} = \frac{1}{n^2} \cdot (1 - 0.5^n)$ by sum of geometric sequence

2a. great answer!

$\hat{\beta}$ is consistent $\Leftrightarrow \lim_{n \rightarrow \infty} \hat{\beta} = \beta$, where $\hat{\beta} = \beta + \frac{1}{n} \sum u_i$

u_i is not identically distributed (heteroskedastic errors) so we can't apply the Chebyshev LLN. However, we can still show consistency. The Chebyshev inequality states

$P(|\hat{\beta} - \beta| > c) \leq \operatorname{var}(\hat{\beta}) / c^2, \quad \forall c > 0 \in \mathbb{R}$

Considering the LHS, $\operatorname{var}(\hat{\beta}) / c^2 = \frac{1 - 0.5^n}{n^2 c^2} \rightarrow \frac{1}{n^2 c^2} \rightarrow 0$ as $n \rightarrow \infty$

so $P(|\hat{\beta} - \beta| > c) \xrightarrow{p} 0$ as $n \rightarrow \infty$, and we have consistency.

b) We're interested in the distribution of $\sqrt{n}(\hat{\beta} - \beta) = \sqrt{n}(\frac{1}{n} \sum u_i) = n^{-1/2} \sum u_i$ ($= \sqrt{n}(\bar{u} - \mu_u)$)

But we can't apply the Lindeberg-Levy CLT here as the errors are not iid (variance changes).

Consider instead the fact that the independent errors u_i are all Normal and mean zero, with $\text{var}(u_i) \rightarrow 0$ as $n \rightarrow \infty$.

So the random variable $\hat{\beta} - \beta = \frac{1}{n} \sum u_i$ is the sample mean of many independent ^{Normal} variables. If we scale by n , then

$$n(\hat{\beta} - \beta) = \sum u_i \text{ will}$$

(i) be normally distributed, since it's the sum of normal variables

(ii) be mean zero, by linearity,

(iii) have limiting variance of 1, since

$$\lim_{n \rightarrow \infty} \text{var}(\sum u_i) = \sum_{i=1}^{\infty} 0.5^i = 1$$

2b. yes, excellent!

$$\text{So } n(\hat{\beta} - \beta) \sim N(0, 1).$$

c) As before, $\hat{\beta}$ will be the (transformed) sample mean.

$$\hat{\beta} = \underset{b \in \mathbb{R}}{\text{argmin}} \sum (y_i \cdot (0.5^i)^{-1/2} - b \cdot (0.5^i)^{-1/2})^2$$

with FOC $\frac{\partial L}{\partial b} = -2 \sum (y_i \cdot (0.5^i)^{-1/2} - b) (0.5^i)^{-1/2} \stackrel{!}{=} 0$

2c. yes, correct!

$$\Rightarrow \hat{\beta} = \frac{\sum y_i / (0.5^i)}{\sum 1 / (0.5^i)} = \frac{\sum 2^i y_i}{\sum 2^i} = \frac{\sum 2^i (\beta + u_i)}{\sum 2^i} = \beta + \frac{\sum 2^i u_i}{\sum 2^i}$$

$$\Rightarrow \hat{\beta} = \frac{\sum y_i / (0.5^i)}{\sum 1 / (0.5^i)} = \frac{\sum 2^i y_i}{\sum 2^i} = \frac{\sum 2^i (\beta + u_i)}{\sum 2^i} = \beta + \frac{\sum 2^i u_i}{\sum 2^i}$$

$$d) E[\tilde{\beta}] = \beta + E\left[\frac{\sum 2^i u_i}{\sum 2^i}\right] = \beta + \frac{1}{\sum 2^i} E\left[\sum 2^i u_i\right]$$

$$= \beta + \frac{1}{\sum 2^i} \sum 2^i E[u_i] \quad \text{by linearity}$$

$$= \beta + \frac{1}{\sum 2^i} \sum 2^i \cdot 0 = \beta \quad \text{since errors mean zero}$$

$$\text{var}(\tilde{\beta}) = \text{var}\left(\frac{\sum 2^i u_i}{\sum 2^i}\right) = \frac{1}{(\sum 2^i)^2} \text{var}(\sum 2^i u_i)$$

$$= \frac{1}{(\sum 2^i)^2} \cdot \sum (2^i)^2 \text{var}(u_i) \quad \text{since } u_i \perp u_j, i \neq j$$

$$= \frac{1}{(\sum 2^i)^2} \cdot \sum 2^{2i} = \frac{1}{(\sum 2^i)^2} \sum_{i=1}^n 2^{2i}$$

$$= \frac{1-2^{2n+1}}{2(1-2^2)} = \frac{1}{2(2^n-1)} \quad \text{by sum of geometric seq.}$$

$$\frac{a(r^{n+1}-1)}{1-r}$$

$$\text{Recall that } \text{var}(\tilde{\beta}) = \frac{1-2^{2n+1}}{n^2} \quad \text{when } n=2,$$

$$\text{var}(\tilde{\beta}) = \frac{1}{6} \quad \text{and} \quad \text{var}(\hat{\beta}) = \frac{0.75}{4} = \frac{3}{16} \quad \text{so } \text{var}(\tilde{\beta}) < \text{var}(\hat{\beta}).$$

And we can see that $\text{var}(\tilde{\beta})$ will shrink faster than $\text{var}(\hat{\beta})$, since it has 2^n in the denominator rather than the slower-growing n^2 , while both numerators $\rightarrow 2$.

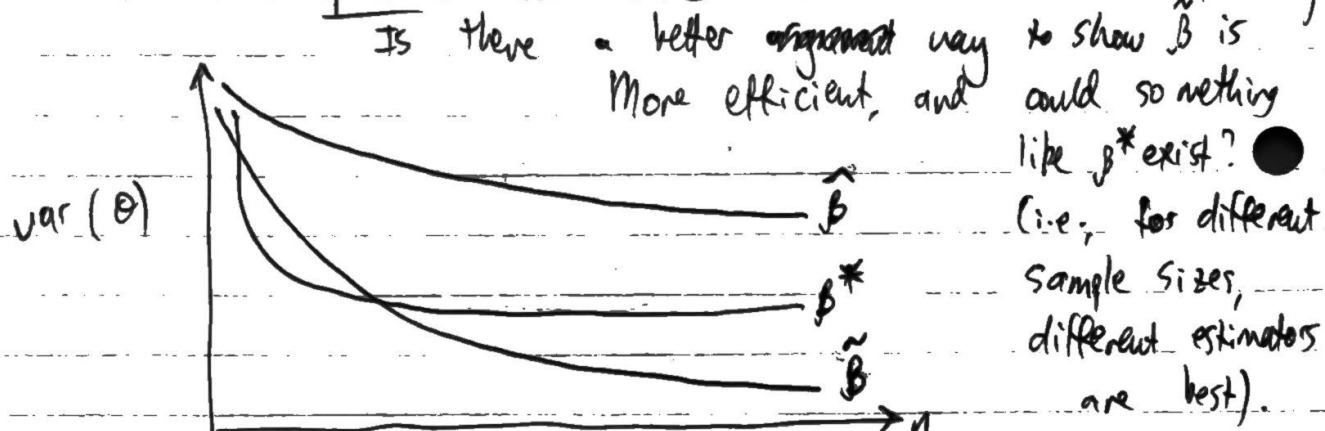
So $\tilde{\beta}$ is more efficient. This makes sense, since after transformation our errors are now homoskedastic with unit variance, so the assumptions for Gauss-Markov are satisfied and hence it is BLUE.

* 1. We can argue $\tilde{\beta}$ is BLUE, but not $\hat{\beta}$. Does this itself imply that $\tilde{\beta}$ is more efficient? The complication is that they're

obtained from two different models, so it's not obvious to me that the "optimality" of $\tilde{\beta}$ should include it being better than estimators for transformed models like $\hat{\beta}$ in (3).

[Maybe this is to do with ^{me} needing better intuitions about what Aitken's theorem means.]

2. My argument about efficiency was quite informal. And also, it seems possible we'd have a situation like the following:



2d. correct derivation of expectation and variance + correct conclusion on the choice of estimator. The informal argument is right acc. to me and is essentially the same logic used in Q1. I wouldn't say the 2 models are different from each other, one is just a transformed version of the other, but they are still the same in that we still have only one regressor. I would check with the tutor for a more definitive answer

the formal proof of why beta tilde is more efficient than beta hat involves proof by induction